NARAYANA IIT ACADEMY INDIA

JEE ADVANCED GRAND TEST SERIES



46 YEARS OF EXCELLENCE

NARAYANA IIT - JEE ACADEMY - INDIA

Sec: OSR.IIT _*CO-SC(MODEL-A&APEX) Date: 22-03-26
Time: 3 Hrs Max. Marks: 198

22-03-26_OSR.IIT_*CO-SC(MODEL-A&APEX)_JEE ADV_GTA-1(P2)_SYLLABUS

Rankers Academy JEE

MATHEMATICS: **TOTAL SYLLABUS**

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

THE NARAYANA GROUP

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TIME: 3HRS **IMPORTANT INSTRUCTIONS** Max Marks: 198

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 6)	Questions with Single digit integer(0-9)	+3	-1	6	18
Sec – II (Q.N : 7 – 12)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 13 – 18)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -1 27.30)	+4	0	6	24
Total				**18**	**66**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 – 24)	Questions with Single digit integer(0-9)	+3	-1	6	18
Sec – II (Q.N : 25 – 30)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 31 – 36)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -1 27.30)	+4	0	6	24
Total				**18**	**66**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 – 42)	Questions with Single digit integer(0-9)	+3	-1	6	18
Sec – II (Q.N : 43 – 48)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 49 – 54)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -1 27.30)	+4	0	6	24
Total				**18**	**66**

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MATHEMATICS **MAX.MARKS: 66**

SECTION – 1 (Maximum Marks: 18)

This section contains SIX (06) questions.
The answer to each question is **A SINGLE DIGIT INTEGER** rangeing from **0 TO 9 , BOTH INCLUSIVE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric
keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

1. Let f be a twice differentiable function with $f(-1)=1=f(1)$ and $f(0)=0$. Then the minimum number of root(s) of equation

$$f(x)\left(f'(x)\right)^3+\left(f(x)\right)^2 f'(x)f''(x)-f(x)f'(x)=x\left(f'(x)\right)^2+xf(x)f''(x)-x$$

is/(are)

2. Let a vector $\overline{V_1}=\overline{OP}$, where O is origin and P is $(2,0,2)$. Vector $\overline{V_1}$ is rotated about origin by an angle of $\frac{\pi}{3}$ in a plane which is perpendicular to XOZ plane to get vector $\overline{V_2}$ where $\overline{V_2}\cdot\hat{j}>0$. Again $\overline{V_1}$ is rotated about origin by right angle in XOZ plane to get vector $\overline{V_3}$. If V is volume of tetrahedron whose coterminous edges are $\overline{V_1},\overline{V_2}$ and $\overline{V_3}$ then value of $\frac{3V}{\sqrt{6}}$ is

3. Suppose that $f(x)$ is a differentiable invertible function with $f(x)\neq 0$ and $h(x)=\int_1^x f(t)dt$. Given that $f(1)=f'(1)=1$ and $g(x)$ is inverse of $f(x)$. Let $G(x)=x^2g(x)-xh(g(x))\forall\in R$ then $G''(1)=$

4. Let $f(x)$ be differentiable function such that $f(x)=x^2+\int_0^x e^{-t}f(x-t)dt$ then $6f(1)=$

5. Number of distinct real solutions of $\sin\pi x=\left|\ln|x|\right|$ is

6. Number of solution(s) of the equation $8000\sin^3 x+3375\cos^3 x+1728=5400\sin 2x$ in $(0,2\pi)$ is (are)

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SECTION - 2 (Maximum Marks : 24)

This section contains SIX (06) questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

7. Each of 2010 boxes in a line contains one red marble and for $1 \le k \le 2010$, the box is the kth position also contain k white marbles. A child begins at the first box and successively drawn a single marble at random from each box in order. The stops when be first draws a red marble. Let $p(n)$ be the probability that he stops after drawing exactly n marbles. The possible value (s) of n for which $p(n) < \frac{1}{2010}$ is

A) 44 B) 45 C) 46 D) 47

8. If $5x\left(1+\frac{1}{x^2+y^2}\right)=12$ and $5y\left(1-\frac{1}{x^2+y^2}\right)=4, x, y \in R$, then correct option is/are

A) $|x|+|y|=3$ B) $x+y=\frac{1}{5}$ C) $|x|+|y|=\frac{3}{5}$ D) $x^2+y^2=5$

9. Let $A_1, A_2 \ldots\ldots A_7$ be a polygon and $a_1, a_2 \ldots\ldots a_7$ be the complex numbers representing vertices $A_1, A_2 \ldots\ldots A_7$. If $|a_1|=|a_2|=\ldots\ldots|a_7|=R$, then $\sum_{1 \le i < j \le 7} |a_i + a_j|^2$

A) greater than $30R^2$

B) has minimum value as $35R^2$

C) has its minimum value in $(25R^2, 45R^2)$

D) is less than $45R^2$

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10. If the sides a,b & c of ΔABC is such that $\frac{a}{1+\alpha^2\beta^2}=\frac{b}{\alpha^2+\beta^2}=\frac{c}{(1-\alpha^2)(1+\beta^2)}$ then

A) $A=2\tan^{-1}\left(\frac{\alpha}{\beta}\right)$ B) $B=2\tan^{-1}(\alpha\beta)$

C) area of $\Delta ABC=\frac{\alpha\beta ab}{\alpha^2+\beta^2}$ D) area of $\Delta ABC=\frac{\alpha\beta bc}{\alpha^2+\beta^2}$

11. If A_i is the area bounded by $|x-a_i|+|y|=b_i, i\in N$, where

$a_{i+1}=a_i+\frac{3}{2}b_i$ & $b_{i+1}=\frac{b_i}{2}, a_1=0, b_1=32$, then

A) $A_3=128$ B) $A_3=256$

C) $\lim\limits_{n\to\infty}\sum\limits_{i=1}^{n}A_i=\frac{8}{3}(32)^2$ D) $\lim\limits_{n\to\infty}\sum\limits_{i=1}^{n}A_i=\frac{4}{3}(16)^2$

12. $\bar{a}$ & $\bar{b}$ be two vectors such that $|a|=1,|b|=4$ & $\bar{a}.\bar{b}=2$. If $\bar{c}=(2\bar{a}\times\bar{b})-3\bar{b}$, then which of following is/are correct?

A) $\bar{b}.\bar{c}=48$ B) $\bar{b}.\bar{c}=-48$

C) Angle between $\bar{b}$ & $\bar{c}$ is $\frac{5\pi}{6}$ D) Angle between $\bar{b}$ & $\bar{c}$ is $\frac{\pi}{6}$

SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

13. An unbiased dice with numbers $\{1,2,3,4,5,6\}$ on its faces is rolled once. An observer, A, sees the number shown on the dice and reports it. The probability that A speaks the truth is $\frac{1}{3}$ and the probability that he lies is $\frac{2}{3}$. Assume that the number he reports is always from the set $\{1,2,3,4,5,6\}$. If p is the probability that the actual number on the dice is even given that A reports that the number was 2, then p is equal to

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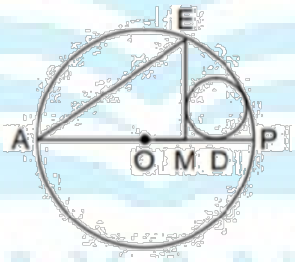
14. If $a_n = \sum_{r=0}^{n} \frac{1}{{}^{n}C_r}$ and $L = \lim_{n\to\infty} \frac{1}{a_n}$, then L is equal to

15. If $S = \sum_{k=1}^{\infty} \frac{1}{{}^{k+51}C_k}$, then S is equal to

16. In ΔABC let the value of $a^3\cos 3B + 3a^2 b\cos(A-2B) + 3ab^2\cos(2A-B) + b^3\cos 3A$ is l, then the value of $\frac{l}{c^3}$ is equal to _____

17. Let $P(x)$ be a polynomial whose coefficient are integer satisfies $P(1)=5$ and $P(2)=7$. The smallest possible positive value of $P(10)$ is equal to _____

18. Let 'O' be centre of circle as shown in figure and AE be a chord from A the diameter AP is drawn and M is foot of perpendicular drawn from E on AP. A circle is drawn such that it touches EM, AP at D and circle whose diameter is AP, then the value of $\frac{AE}{AD}$ is _____



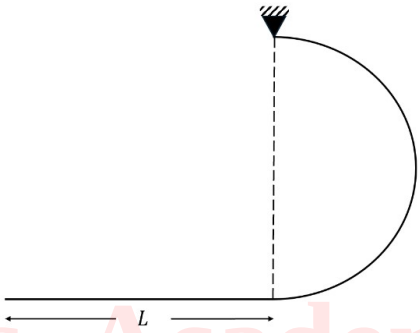
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PHYSICS **MAX.MARKS: 66**

SECTION – 1 (Maximum Marks: 18)

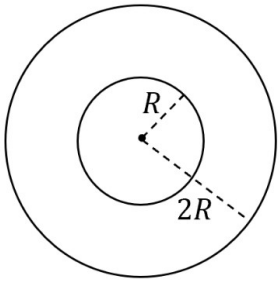
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19. A thin uniform rod of length L is rigidly attached to a thin uniform semi-circular ring of same linear density to form a rigid structure as shown. The system is hinged as shown and at equilibrium the rod is horizontal. The distance of the center of mass of the system from the hinge is then $\frac{L}{2}\left(\frac{\pi+\alpha}{\pi+\beta}\right)$. Find $\alpha\beta$.



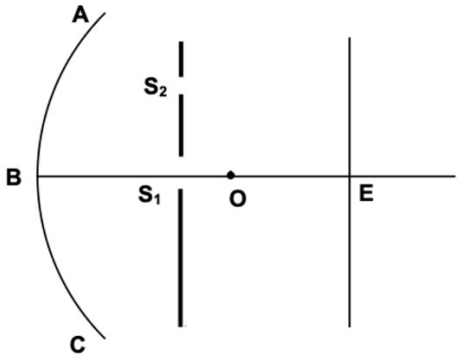
20. A cubical container (material of thermal conductivity K) of side a and wall thickness $x(x << a)$ is suspended in air and filled with n moles of a diatomic gas (adiabatic exponent γ) in a room of room temperature T_0. If at time $t=0$, the temperature of the diatomic gas is $T_1(T_1 > T_0)$, then the temperature of the gas at a time t is given by $T = pT_0 + (T_1 - T_0)\exp\left[-\frac{qKa^2(\gamma+s)t}{nRx}\right]$. Find $p+q+s$.

21. Two bubbles contain the same ideal gas at the same temperature such that one is trapped inside the other. The system is kept in vacuum. Ratio of densities of the gas in the inner and outer bubbles is x and the ratio of the number of moles of the gas in the inner and outer bubbles is y. Find $\frac{x}{y}$.

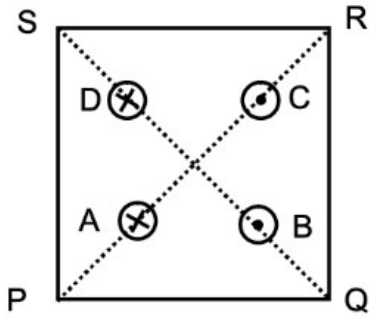


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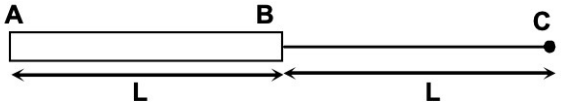
22. ABC is a spherical wavefront centered at O. It is incident on slits S_1 and S_2 such that $BS_1 = 3\lambda, S_1S_2 = 4\lambda; BO = 6\lambda$ and $S_1E = 128\lambda$, where λ is the wavelength of the incident wave. A mica sheet of refractive index 1.5 is placed in front of S_2. The minimum value of the thickness of the mica sheet such that the central maximum is formed at E is $\frac{(10k+1)\lambda}{8}$. Find the value of k.



23. $PQRS$ is a square of side l_0. A, B, C and D are four long wires carrying currents perpendicular to the plane of the paper as shown. $PA = QB = RC = SD = \frac{l_0}{\sqrt{5}}$. The magnitudes of the currents in A, B, C and D are $i_0, 2i_0, 8i_0$ and $4i_0$ respectively. It is given that $\int_P^Q \vec{B}_A \cdot \vec{dl} = -9\mu_0$ Tesla meter and $\int_R^S \vec{B}_B \cdot \vec{dl} = 5\mu_0$ Tesla meter, where $\vec{B}_A$ and $\vec{B}_B$ are the magnetic fields due to the wires A and B respectively. The magnitude of i_0 is $3p+2$. Find the value of p.



24. A uniform rod AB of mass M length L is fixed. A point mass is released at rest from point C as shown in the figure. The speed of the mass is found to be $v = \sqrt{\frac{xGM}{L}\ln\left(\frac{y}{2}\right)}$ when the mass is at a distance of $\frac{L}{2}$ for the end B of the rod. Find $x + y$.



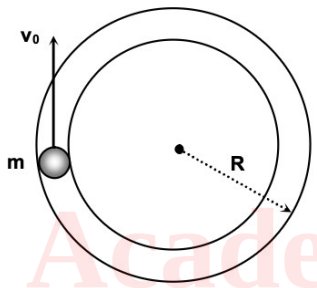
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SECTION - 2 (Maximum Marks : 24)

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Negative Marks: -2 In all other cases.

25. A small ball of mass m is placed in a circular tube of negligible width and mass M and radius R which is kept on a smooth horizontal plane in a gravity free space. Friction is absent between the ball and the tube. The ball is given a velocity v_0 as shown. Choose the correct option(s):



A) The path of the ball relative to the center of mass of the (ball + tube) system is circular.

B) The path of the ball relative to the center of mass of the (ball + tube) system is elliptical

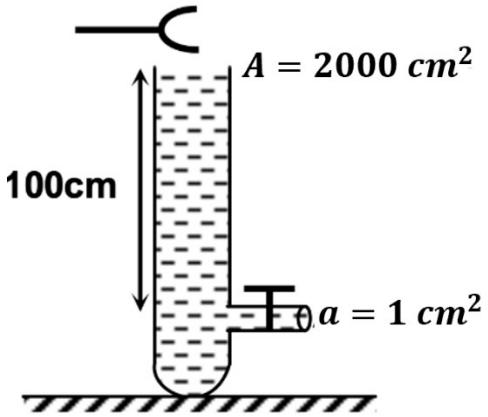
C) Radius of curvature of the trajectory of the ball relative to the ground at the time of projection is $\frac{MR}{m+M}$

D) Normal force between the tube and the wall if $M=2m$, at the time of projection of the ball is $\frac{2mv_0^2}{3R}$.

26. A tuning fork of frequency $1000Hz$ is sounded over a resonance tube as shown. Speed of sound is $320\,m/s$ and the area of cross section of the tube is $A=2000\,cm^2$. At a depth of $100\,cm$ a water outlet is attached which has an cross-sectional area of $a=1cm^2$ and is opened at $t=0\,s$. Neglect end correction.

(Take $g=10m/s^{-2};\sqrt{0.92}=0.96;\sqrt{0.76}=0.87$)

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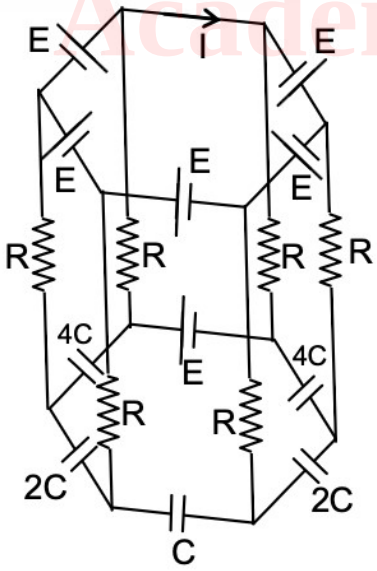
A) First resonance will occur when the water level goes down by $8\,cm$ from the open end.

B) First resonance will occur when the water level goes down by $16\,cm$ from the open end.

C) The time at which the first resonance occurs is nearly at $t = 0.6\,\text{min}$

D) The time difference between the first two resonances is nearly $\Delta t = 1.3\,\text{min}$

27. A hexagonal prism wire frame is made with 6 sources of equal emf $\varepsilon = 5V$ and internal resistance $r = 1\Omega$. Six equal resistances each equal to $R = 2\Omega$ and four capacitors of equal capacitance $C = 2\,\mu F$ are connected as shown. The correct statement(s) are



A) Value of $I = 6\,A$

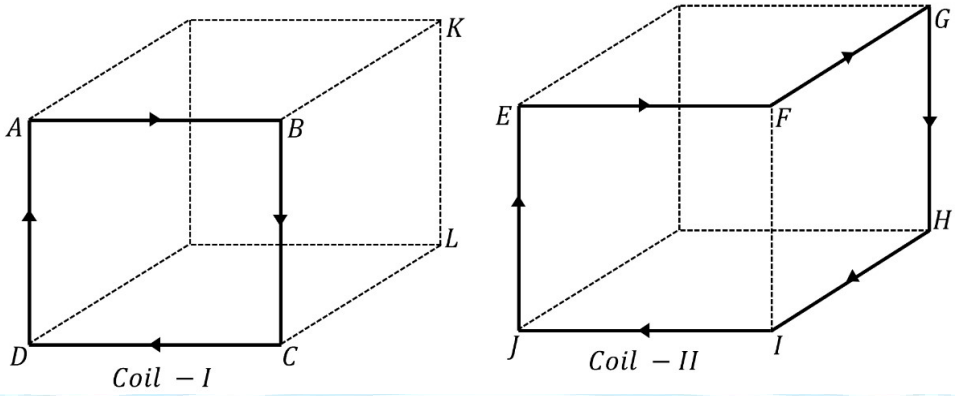
B) Value of $I = 5\,A$

C) Value of charge on the $4C$ capacitor is $16\,\mu C$

D) The value of charge on the capacitor with capacitance C is zero

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28. A conducting *Coil – I* named $ABCD$, shown as a part of an imaginary cube, has a self-induction coefficient L_1. Another conducting *Coil – II* named $EFGHIJ$, shown as a part of an identical imaginary cube as *Coil – I*, has a self-induction coefficient L_2. $BKLC$ is a neighboring surface of *Coil – I*. Identify the correct statement(s): (ignore mutual induction between the coils)



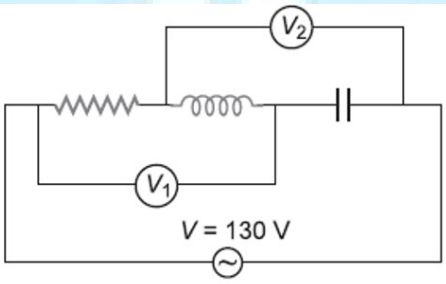
A) If a current i flows through *Coil – I* in the indicated direction, the self-induction flux through *Coil – I* is $L_1 i$

B) If a current i flows through *Coil – II* in the indicated direction, the self-induction flux through *Coil – II* is $L_2 i$

C) If a current i flows through *Coil – I* in the indicated direction, the flux through the neighboring surface $BKLC$ is $\frac{L_2}{2} i$

D) If a current i flows through *Coil – I* in the indicated direction, the flux through the neighboring surface $BKLC$ is $\left(\frac{L_2}{2} - L_1\right) i$

29. In a R-L-C series circuit shown, the readings of voltmeters V_1 and V_2 are $50\sqrt{2}\,V$ and $120\,V$ respectively. Choose the correct statement(s)



A) Voltage across resistor, inductor and capacitor are 50V, 50V and 170V respectively.

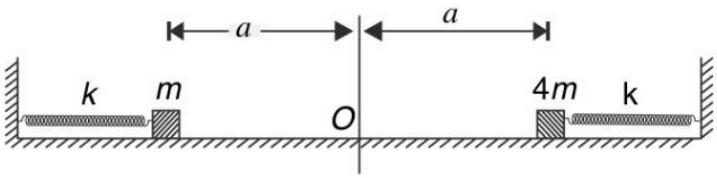
B) Voltage across resistor, inductor and capacitor are 10V, 90V and 30V respectively.

C) power factor of the circuit is $\frac{5}{13}$

D) circuit is capacitive in nature.

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30. Two small blocks of mass m and $4m$ are connected to two springs as shown in figure. Both springs have stiffness K and they are in their natural length when the blocks are at point O. Both the blocks are pushed so that both the springs get compressed by a distance a. First the block of mass m is released and after it travels a distance $\left(1-\frac{\sqrt{3}}{2}\right)a$, the second block is also released.



A) The distance from point O where the two blocks first collide is $a\cos\left(\frac{5\pi}{18}\right)$

B) The distance from point O where the two blocks first collide is $a\cos\left(\frac{5\pi}{9}\right)$

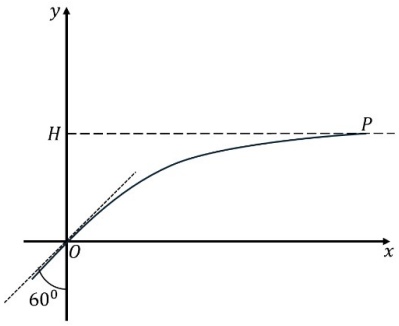
C) The time at which the two blocks collide after the block of mass $4m$ is released is $\frac{5\pi}{9}\sqrt{\frac{m}{k}}$

D) The time at which the two blocks collide after the block of mass $4m$ is released is $\frac{5\pi}{18}\sqrt{\frac{m}{k}}$

SECTION - 3 (Maximum Marks : 24)

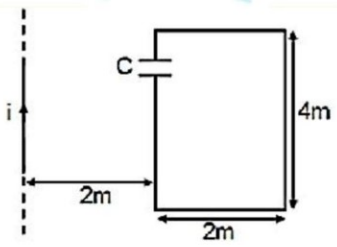
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Zero Marks: 0 In all other cases.

31. A system of coordinates is drawn in a medium whose refractive index varies according to $\mu=\frac{2}{1+y^2};0<y\le 1$ and $\mu=1;y>1$, A ray of light is incident at the origin making an angle of 60^0 with the $y-$axis as shown. At point P the ray becomes parallel to the $x-$axis as shown at $y=H$. Find $\frac{1}{H^2}$. (Take $\sqrt{3}=1.73$)

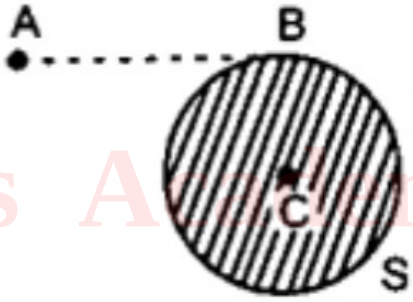


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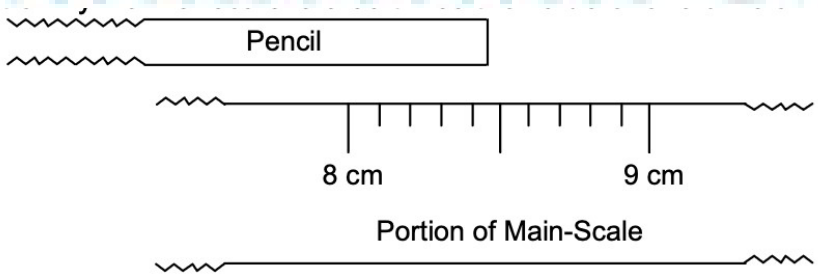
32. A rectangular frame of negligible resistance with capacitor of capacitance $C=5\mu F$ is placed near an infinitely long wire carrying current $I=I_0\sin\left(\omega t+\frac{\pi}{6}\right)A$ with the wire and the frame being in the same plane. Charge on the capacitor (in μC) at $t=0s$ is $\left(\omega=10^4\ rad/s, i_0=10^2\ A, \sqrt{3}=1.73, \ln(2)=0.693\right)$



33. S is a neutral conducting sphere. A point charge $q=1\mu C$ is placed at A. C is the center of the sphere and AB is a tangent. $BC=3m$ and $AB=4m$. The electric potential at B due to the induced charges is $-xkV$. Find x.

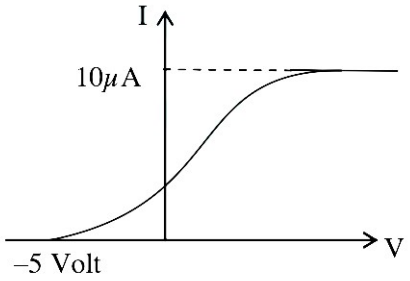


34. The length of a pencil is being measured by a Vernier caliper. The least count is 0.05 times the value of one division of the main scale. The Vernier scale has n divisions which coincide with $(n-4)$ divisions of the main scale. Also, while taking the reading it is noticed that the $\left(\frac{n}{5}\right)^{th}$ division of the Vernier scale exactly coincides with one of the main scale divisions. The length of the pencil (in cm) is:
(assume there is no zero error)

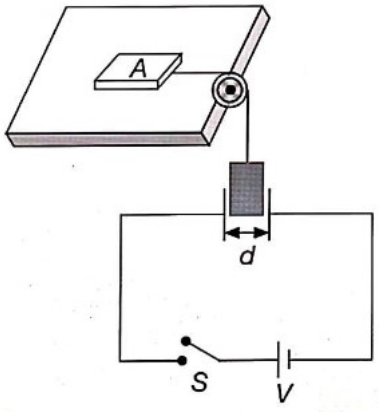


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35. In the photo-electric experiment, if we use monochromatic light, the I-V curve is as shown. If the work function of the metal is $2.35eV$, estimate the power of light used (in W) given the efficiency of photoemission $10^{-3}\%$.



36. Block A of mass m is kept on a rough horizontal surface and is connected to a dielectric slab of mass $\frac{m}{6}$ and dielectric constant k by means of a light inextensible string passing over a fixed pulley as shown. The dielectric can completely fill the space between the capacitor plates of area $l \times l$ and separation between the plates d kept in vertical position. Initially the switch is open and the length of the dielectric inside the capacitor is b. The coefficient of friction between block A and the surface is $\frac{1}{4}$. The minimum value of emf V of the battery so that just after closing the switch block A begins to move is $V_{\min} = \sqrt{\frac{mgd}{x\varepsilon_0 l(k-1)}}$. If $V = 2V_{\min}$, then the speed of block A when the dielectric completely fills the space between the capacitor is $\sqrt{yg(l-b)}$. Find xy.



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CHEMISTRY **MAX.MARKS: 66**

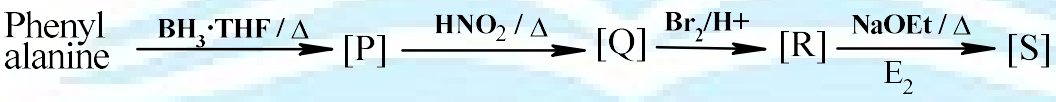
SECTION – 1 (Maximum Marks: 18)

This section contains SIX (06) questions.
The answer to each question is **A SINGLE DIGIT INTEGER** rangeing from **0 TO 9 , BOTH INCLUSIVE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric
keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

37. The primary alcohol 2-methoxyethanol $CH_3O\text{-}CH_2\text{-}CH_2\text{-}OH$, can be oxidized to the corresponding carboxylic acid with aqueous nitric acid (HNO_3). The by-product of the oxidation is nitric oxide, NO. How many moles of HNO_3 are required to oxidize 4.5 moles of alcohol?

38. A mixture of He and O_2 is used to inflate a balloon to a volume of 50L at 1atm and 27^0C. The balloon just barely rises from the table. The balloon weighs 8g and the molar mass of air is 29g. The percentage composition of He in the gas mixture used to fill balloon by weight is? (nearest integer)

39. How many of the following statements is/are **CORRECT** for the following sequence of reactions?



i) The formation of [R] from [Q] is electrophilic substitution reaction

ii) The conversion of [P] to [Q] involves methyl shift

iii) The number of π-bonds in [S] > [Q] > [P]

iv) The compound [Q] can undergo self-aldol condensation

v) The compound [Q] gives yellow precipitate with sodium hypoiodite

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40. A saturated calomel electrode is coupled through a salt bridge with a quinhydrone electrode placed in a solution containing a weak acid HA at 0.02M and NaA at 0.04M. The observed EMF at 25^0 C is 0.158V. Given E^0 of saturated calomel is 0.242V and quinhydrone electrode is 0.70V. If the dissociation constant of HA is $P\times10^{-Q}$, then the value of (P + Q) is? $\left(\frac{2.303RT}{F}=0.06\right)$

41. White phosphorus reacts with thionyl chloride and form compounds A, B and C. Chlorine is present in compounds A and B while sulphur is present in compounds B and C. If percentage by mass of sulphur is more in C than B, then how many of the following is/are **INCORRECT**?

i) Compound A is used in vulcanization of rubber

ii) Compound B is used in the conversion of alcohols into alkyl chlorides

iii) Total number of lone pairs per molecule of C is greater than B

iv) The hybridization of central atoms in A and B is same but in C is different

v) Same products are formed when white phosphorus reacts with sulfuryl chloride

vi) The number of lone pairs on the central atom of A and C is same

42. How many of the following bond length orders is/are **CORRECT**?

(a) Bond length N_2^+ > Bond length N_2

(b) Bond length NO^+ < Bond length of NO

(c) Bond length CN^- < Bond length of CN

(d) Bond length O_2^- < Bond length of O_2^{-2}

(e) Bond length O_2 > Bond length of O_2^+

(f) Bond length B_2 > Bond length of B_2^-

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SECTION - 2 (Maximum Marks : 24)

This section contains SIX (06) questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

43. Which of the following statements is/are **CORRECT**?

A) When sulphur dioxide gas is bubbled through an aqueous solution of copper sulphate in presence of potassium thiocyanate, gives cuprous thiocyanate.

B) Phosphorus can react with nitric acid of appropriate concentration to give equimolar ratio of nitric oxide and nitrogen dioxide.

C) Sodium iodate on treating with sodium bisulphite(not in excess) solution, liberates iodine.

D) Anhydrous potassium nitrate when heated with excess of metallic potassium gives nitrogen dioxide

44. Analyze the following data and choose the **CORRECT** option(s)

Compound [X]	**Compound [Y]**
M.F: C_4H_6O (acyclic compound)	M.F: C_4H_6O (acyclic compound)
Gives positive tollens test	Gives positive tollens test
Gives positive bromine water test	Gives positive bromine water test
Can show tautomerism	Cannot show tautomerism

A) [X] & [Y] are positional isomers

B) [Y] is an aldol condensation product of acetaldehyde

C) The number of replaceable hydrogens with deuterium, when [X] is treated with D_3O^+ is 4

D) [X] & [Y] are chain isomers

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45. For the first order reaction, $2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$

A) the concentration of the reactant decreases exponentially with time.

B) the half–life of the reaction decreases with increasing temperature.

C) the half–life of the reaction depends on the initial concentration of the reactant.

D) the reactant proceeds to almost 99.6% completion in eight half–lives

46. Which of the following statement(s) is/are **CORRECT**?

A) Excess of lithium makes LiCl crystals pink due to F – centres.

B) When silicon is doped with group-15 element n-type semiconductor is formed.

C) H_2O, NaCl and C_6H_6 are some examples of diamagnetic substances.

D) AgBr shows both schottky and frenkel defects..

47. Which of the following statements is/are **CORRECT**?

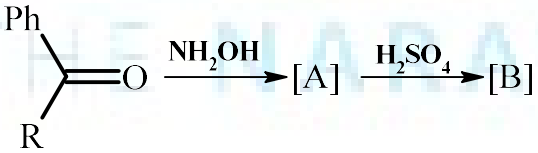
A) The contraction in transition series is attributed to the imperfect shielding of one electron by another in the same subshell

B) The shielding of one f-electron by another is less than that of one d-electron by another with the increase in nuclear charge along the series in the same subshell

C) The size of Ln^{+3} ions smoothly increases as their atomic number increases.

D) As a result of Lanthanoid contraction, Zr and Hf have almost identical radii which accounts for their occurrence together in nature and for difficulty faced in their separation

48. Which of the following is **correct** about the reaction:



A) 'A' consists of mixture of oximes, which undergo rearrangement with the same rate.

B) The rate of rearrangement of 'A' to 'B' is slow in D_2O than in H_2O.

C) When H_2O^{18} is used in the rearrangement step, the amide is found to contain O^{18}.

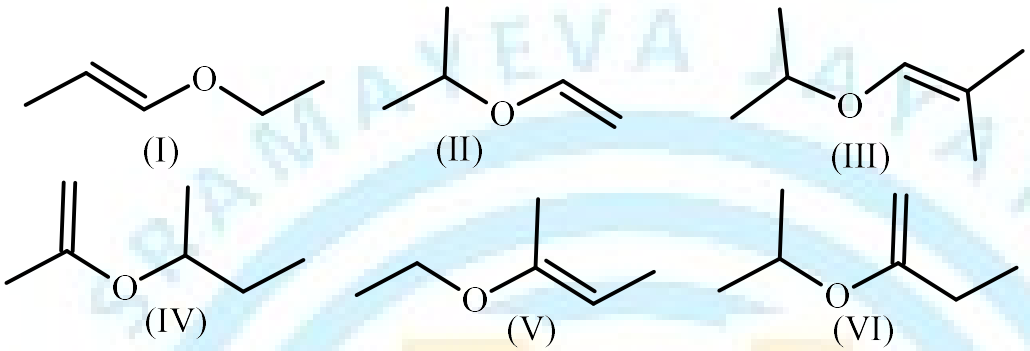
D) 'A' consists of two oximes, both undergo rearrangement at different rates

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SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

49. In how many of the following ethers, both the hydrolysed products undergo +ve iodoform test?



50. 3g of 98% pure sample of $K_xH(C_2O_4)_y$ is reacted with 100 ml of a 0.1M acidified $KMnO_4$ solution. The excess oxidant required 20ml of 0.5M Mohr salt solution for complete reaction. The same mass of another sample of $K_xH(C_2O_4)_y$ is reacted with 0.4M NaOH. The volume of NaOH required for complete reaction in ml is?

51. Loratadine, marketed as an antiallergy medication under the name Claritin, has four rings, eight double bonds and the formula $C_{22}H_nClN_2O_2$. How many hydrogen atoms are present per molecule of loratadine?

52. ΔG^0 for the reaction, $CO_{(g)} + Cl_{2(g)} \rightleftharpoons COCl_{2(g)}$ in cal mol^{-1} is given by,

$\Delta G^0 = -24.5x10^3 + 4TlnT + 3.5T$. Assuming ideal behaviour, the magnitude of ΔH^0 for the reaction at 400K in kcal mol^{-1} is? (nearest integer)

53. The enthalpy of hydration of Fe^{+2} ion is 12 Kcal mol^{-1} higher than what would be expected if there were no crystal field stabilisation energy. The magnitude of Δ_0 in kcal for $[Fe(H_2O)_6]^{2+}$ is?

54. A 1.60-g sample of a mixture of naphthalene ($C_{10}H_8$) and anthracene ($C_{14}H_{10}$) is dissolved in 20.0 g benzene (C_6H_6). The freezing point of the solution is 2.81^0C. The mass percent of naphthalene in the mixture is? The freezing point of benzene is 5.51^0C and K_f is 5.12^0C kg/mol.

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